Reinforcement Learning for Embodied AI: Algorithms, Tasks, and Use Cases

Course home · 简体中文 · PDF

Item	Details
Applicability	Algorithm and task taxonomy; no single framework environment is prescribed
Documentation review date	2026-10-18
License	CC BY 4.0 · Xiong Zeng and contributors

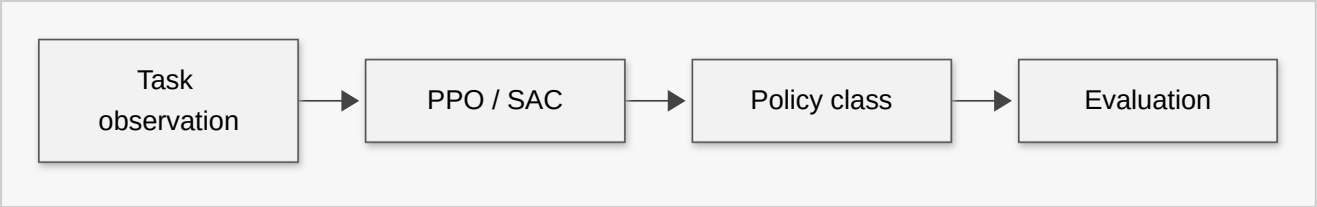
Verification scope: Documentation organization and upstream reference snapshot; no tutorial runtime, training or hardware validation. See version notes for pinned source references. Original teaching content uses CC BY 4.0; third-party quotations, figures and code retain their original licenses. Citation alone does not grant relicensing rights.

Contents

- 1. Overall Classification of Embodied Reinforcement Learning
  - 1.1 Classification Framework
  - 1.2 Core Classification Table
- 2. Classification by Algorithm: PPO and SAC
  - 2.1 On-policy RL: PPO
  - 2.2 Off-policy RL: SAC
  - 2.3 Comparing PPO and SAC
- 3. Classification by Task: Robot Motion-Control Reinforcement Learning
  - 3.1 Locomotion RL: Robot Motion Control
  - 3.2 Manipulation RL: Robot Manipulation Reinforcement Learning
- 4. Classification by Task: Embodied Foundation-Model RL Post-Training
  - 4.1 What Is Foundation-Model RL Post-Training?
  - 4.2 What Problems Does RL Post-Training Solve?
  - 4.3 Five Types of Foundation-Model RL Post-Training
  - 4.4 Summary of the Five Types
- 5. How PPO and SAC Are Used Across the Two Task Categories
  - 5.1 PPO + Locomotion
  - 5.2 SAC + Manipulation
  - 5.3 PPO-Style Methods + VLA
  - 5.4 SAC / Off-policy Critic-Style Methods + VLA
- 6. How to Choose an RL Method for Your Task
- 7. Typical Experimental Paths for the Two Research Directions

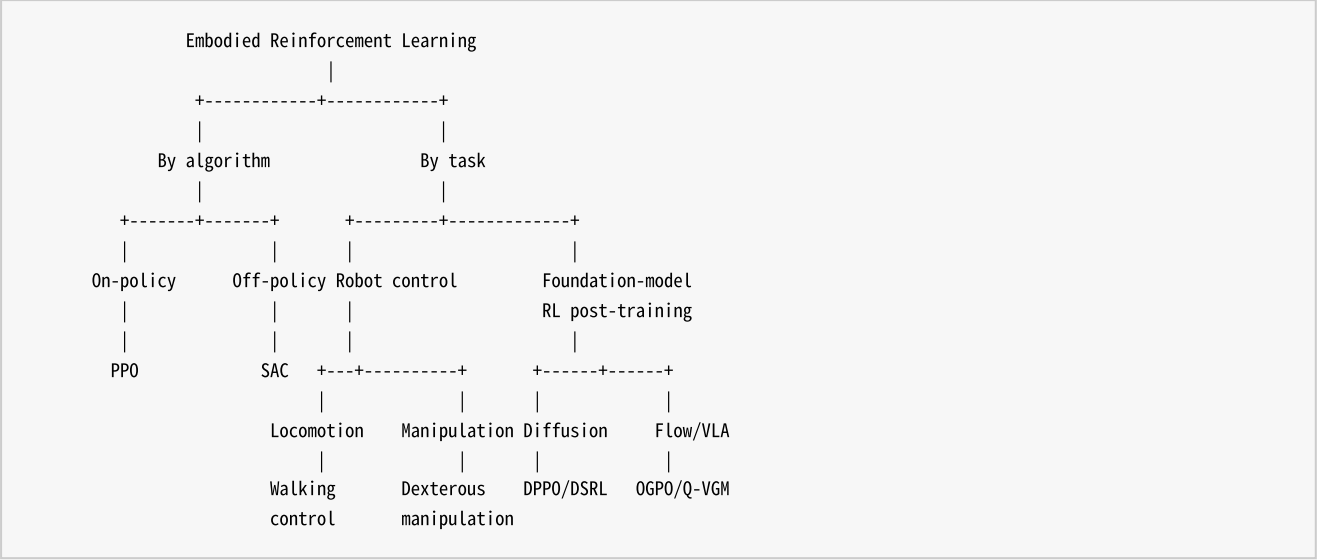
    - 7.1 Robot Motion-Control RL
    - 7.2 Embodied Foundation-Model RL Post-Training
- 8. Final Summary

Chapter concept map



1. Overall Classification of Embodied Reinforcement Learning

1.1 Classification Framework



Robot manipulation can employ either conventional low-level control policies or Diffusion Policies and VLAs.

Therefore, **the same manipulation task can fall under conventional robot-control RL or foundation-model RL post-training; what matters is the policy model and training objective.**

1.2 Core Classification Table

Classification dimension	Category	Representative algorithm or method	Main use
Algorithm	On-policy	PPO	Large-scale simulation training and motion control
Algorithm	Off-policy	SAC	Continuous control, data-efficient learning, and online improvement
Task	Locomotion RL	PPO, SAC	Bipedal, quadrupedal, and humanoid movement

Classification dimension	Category	Representative algorithm or method	Main use
Task	Manipulation RL	PPO, SAC	Grasping, in-hand manipulation, and dexterous control
Task	Foundation Model RL	DPPO, DSRL, OGPO, etc.	Improve pretrained Diffusion / Flow / VLA policies

Note that DPPO, DSRL, and OGPO were developed for generative policies and cannot all be classified directly as standard PPO or SAC.

---

2. Classification by Algorithm: PPO and SAC

2.1 On-policy RL: PPO

PPO is an on-policy optimizer using fresh rollouts and advantage estimates to control policy updates. Here the emphasis is when it is appropriate for parallel simulation and robot tasks. **For clipped surrogate objectives, GAE and full derivation, use the dedicated PPO chapter.**

2.2 Off-policy RL: SAC

SAC reuses replayed interaction data and optimizes a maximum-entropy continuous-action objective with twin critics. This overview focuses on task selection and sample efficiency. **Bellman targets, entropy tuning and Actor/Critic derivations are owned by the SAC chapter.** Applying SAC to a generative VLA is not a trivial actor replacement.

2.3 Comparing PPO and SAC

Dimension	PPO	SAC
Type	On-policy	Off-policy
Training data	Primarily the current policy	Current and historical policy data
Reuse of old data	Limited	More extensive
Replay buffer	Usually no long-term replay buffer	Required
Sample efficiency	Typically lower	Typically higher
Parallel simulation	Very mature ecosystem	Also supported
Training stability	Often easier to manage	Depends on critic quality
High-cost physical interaction	Usually less advantageous	Often more advantageous
Typical tasks	Locomotion	Manipulation, online RL
Representative framework	RSL-RL	Stable-Baselines3 and other SAC implementations

How should you choose?

Choose PPO when:

Robot simulation is fast; a large number of environments can run in parallel; and the goal is to train a stable motion-control policy from scratch.

Choose SAC when:

Physical interaction data are expensive; historical experience needs to be reused; and the goal is continued online policy improvement.

In practice, the choice also depends on the policy architecture. If the policy is a large Diffusion or Flow Matching model, standard PPO/SAC cannot simply be applied without modifications.

---

3. Classification by Task: Robot Motion-Control Reinforcement Learning

The main objective of robot motion-control RL is:

To enable a robot to learn, through interaction with its environment, to execute specific physical movements stably and efficiently.

Usually, a relatively lightweight control policy is trained directly and then deployed to the robot's control system.

3.1 Locomotion RL: Robot Motion Control

Locomotion RL mainly targets robot mobility and whole-body movement capabilities.

Typical applications

Area	Task	Main use
Quadruped Locomotion	Quadrupedal walking, running, and jumping	Robot-dog motion control
Humanoid Locomotion	Bipedal walking and stair climbing	Humanoid locomotion
Motion Tracking	Tracking human or reference motions	Humanoid motion imitation
Whole-Body Control	Coordinated whole-body movement	Mobile manipulation and complex whole-body actions
Push Recovery	Recovery from external pushes	Improve robot stability
Terrain Adaptation	Slopes, gravel, and stairs	Adaptation to challenging terrain

Typical training process

```
Robot model
    |
    v
Build a physics simulation environment
    |
    v
Define the motion task and rewards
    |
    v
Train with PPO
    |
    v
Domain Randomization
    |
    v
Sim-to-Real
    |
    v
Deploy on a physical robot
```

What are the major research topics?

1. Motion capabilities

Enable robots to perform increasingly complex motion tasks, for example:

- Progress from flat-ground walking to stairs;
- Progress from slow walking to running;
- Progress from simple gaits to whole-body motion tracking.

2. Robustness

Improve how well robots cope with different environments, for example:

- External pushes;
- Friction changes;
- Unknown payloads;
- Actuator errors.

3. Sim-to-Real

Reduce the difference between simulated and physical robots.

Main techniques include:

- Domain Randomization;
- System Identification;
- Teacher-Student Learning;
- Privileged Learning;
- Online Adaptation.

Typical technical stack

Module	Common choice
Simulation	Isaac Lab, MuJoCo
RL	PPO
Implementation	RSL-RL
Control policy	MLP, RNN
Low-level control	PD / Torque Control
Training approach	Large-scale parallel simulation
Deployment	Real-time robot control system

Defining feature of this direction:

It is often possible to train from scratch in simulation. The main bottlenecks are physics simulation, reward design, task complexity, and robustness after deployment on physical hardware.

3.2 Manipulation RL: Robot Manipulation Reinforcement Learning

Manipulation RL mainly addresses robot-arm and dexterous-hand operation.

Although it and locomotion RL are both forms of robot control RL, manipulation tasks often involve more complicated contact dynamics.

Typical applications

Area	Task	Main use
Reaching	Move a robot arm to a target	Basic end-effector control
Grasping	Grasp different objects	Robotic grasping
In-hand Manipulation	In-hand rotation and repositioning	Fine dexterous-hand manipulation
Object Reorientation	Adjust object orientation	Dexterous manipulation
Tool Manipulation	Use tools such as screwdrivers	Tool operation
Contact-Rich Manipulation	Insertion, rotation, assembly	Precision contact operations
Bimanual Manipulation	Coordinated two-arm control	Two-handed manipulation and assembly

Differences from locomotion RL

Locomotion focuses more on:

- Body balance;
- Velocity tracking;
- Ground contact;
- Periodic movement.

Manipulation focuses more on:

- Multiple contact points;
- Object pose;
- Grasp stability;
- Contact-mode transitions;
- Finger coordination;
- Precision operation.

Should you use PPO or SAC?

PPO:

Suitable for training many dexterous manipulation tasks in parallel simulation.

For example, train a dexterous hand to rotate objects from different initial object poses.

SAC:

Suitable for research on real-robot online learning and manipulation tasks requiring high environment-interaction sample efficiency.

For example, start from an existing manipulation policy and use a small number of real-world interactions to progressively improve grasp success.

Important distinction

Here, the trained policy may still be a conventional low-level controller:

```
Observation
    |
    v
MLP Actor
    |
    v
Joint Position / Torque
    |
    v
Robot
```

This is a different problem setting from VLA reinforcement-learning post-training in the next section.

---

4. Classification by Task: Embodied Foundation-Model RL Post-Training

4.1 What Is Foundation-Model RL Post-Training?

Embodied foundation models are usually pretrained on large amounts of demonstration data and then improved through reinforcement learning.

Examples include:

- Diffusion Policy;
- Flow Matching Policy;
- VLAs such as π0 and π0.5;
- Other vision-language-action generative models.

The typical training process is:

```
          Large-scale robot demonstrations
                        |
                        v
                Imitation Learning
                        |
                        v
                 Pretrained Policy
                        |
                        v
                 RL Post-training
                        |
                        v
                 Improved Robot Policy
```

The main difference from training a locomotion policy from scratch is that:

The post-training stage begins with a pretrained policy that already has some task capability.

The purpose of RL is no longer to learn basic movements from scratch, but to further improve performance based on existing capabilities.

4.2 What Problems Does RL Post-Training Solve?

Problem	Role of RL post-training
Insufficient success rate	Improve success at grasping, placing, insertion, and related tasks
Slow execution	Reduce task-completion time
Insufficient precision	Improve end-effector positioning and precision manipulation
Failure in new environments	Adapt online to environmental changes
Performance limited by imitation learning	Use autonomous experience to exceed demonstration performance
No recovery after failure	Learn recovery and retry behaviors
Few demonstrations for real tasks	Improve further through autonomous exploration

For example:

A pretrained VLA can already grasp a screwdriver, but its success rate is only 70%.

We would like to use reinforcement learning to improve success rate, eliminate unnecessary movements, and increase robustness to variations in initial pose.

4.3 Five Types of Foundation-Model RL Post-Training

These methods can be divided into five categories according to **which part of the model reinforcement learning updates.**

Type 1: Full Policy Fine-tuning

Directly update the pretrained generative policy.

Representative methods:

- DPPO;
- OGPO.

DPPO uses policy gradients to fine-tune Diffusion Policies. OGPO uses an off-policy critic and an optimization objective adapted to the generation process to perform full fine-tuning of generative control policies. (DPPO)

Main use:

Full policy fine-tuning is appropriate when a pretrained policy's capabilities are clearly insufficient and its action-generation behavior needs to change substantially.

The advantage is a broader scope for modifying the policy and potentially improving its capabilities.

The drawbacks are high training costs and the need to avoid damaging previously learned capabilities.

Type 2: Latent / Noise-Space RL

Freeze the pretrained model and optimize only its input latent variables or noise distribution.

Representative methods:

- DSRL;
- LP-DS.

DSRL improves a Diffusion Policy by learning a distribution over its initial noise, without updating the parameters of the base policy. (DSRL)

Main use:

When a pretrained model already generates fairly good actions and the main objective is to select better modes of behavior.

Advantages:

- No need to fine-tune the entire large model;
- Relatively lightweight online training;
- Reuse of the pretrained policy.

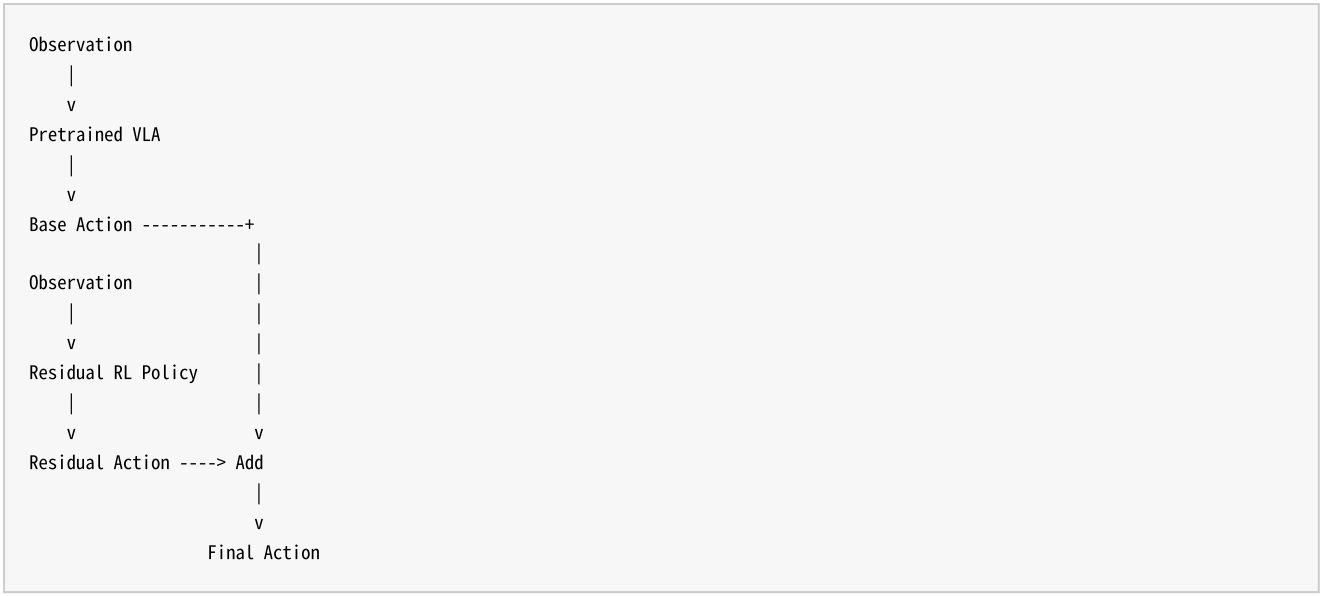
Limitation:

It is difficult to go beyond the representational capabilities of the frozen generator itself.

Type 3: Action Residual RL

Add RL-based corrections to the actions produced by a pretrained policy.

For example:



Main use:

The pretrained policy is mostly correct, but end-effector errors, pose errors, or contact control are not accurate enough.

For example:

- Increase insertion accuracy;
- Correct grasp pose;
- Add local control for tool operations;
- Improve actions during contact phases.

This category usually does not require retraining the base VLA.

Type 4: Feature / Token-Level RL

Train a lightweight RL module using internal features of a pretrained VLA.

Representative method:

- RL Token.

RL Token uses the pretrained VLA to provide compact task-relevant representations and trains a lightweight actor-critic module for online policy improvement. (RL Token)

Main use:

Online reinforcement learning for large VLAs.

Examples:

- Precision assembly;

- Insertion;
- Tool operation;
- Improving execution speed online.

This category is appropriate when the goal is to preserve the foundation model's representations while reducing online training costs.

Type 5: Velocity Field Fine-tuning

Directly optimize the velocity field of a Flow Matching Action Expert.

Representative research:

- Q-VGM;
- Approaches related to Reward / Advantage-Guided Velocity Matching.

Q-VGM uses value-gradient signals provided by a critic to improve a pretrained Flow Matching Policy through local Velocity Matching. (Q-VGM)

Main use:

When the goal is to genuinely improve a Flow Matching action-generation model without expensive backpropagation through the entire generative trajectory for every policy update.

This is a research direction highly relevant to post-training large Flow Matching VLAs.

4.4 Summary of the Five Types

Method	Representative	Updated component	Main use
Full Fine-tuning	DPPO, OGPO	Entire generative policy or main Action Expert	Improve the model's action-generation capabilities
Latent RL	DSRL, LP-DS	Input noise or latent policy	Lightweight online improvement
Action Residual RL	Residual SAC, etc.	Residual of the output action	Precision control and local action correction
Feature / Token RL	RL Token	Lightweight actor-critic module	Efficient online learning for large VLAs
Velocity Field RL	Q-VGM, etc.	Flow Matching velocity field	Directly improve the Flow Matching Action Expert

The first four categories mainly differ in where parameters are optimized; the fifth additionally exploits the generative structure of Flow Matching.

These categories are not completely mutually exclusive. For example, training only a velocity-field residual module falls under both residual adaptation and velocity-field fine-tuning.

---

5. How PPO and SAC Are Used Across the Two Task Categories

Combining the two classification dimensions gives the following core table.

	Robot motion control	Foundation-model RL post-training
On-policy PPO	Train locomotion control policies in large-scale simulation	Generative-policy fine-tuning such as DPPO
Off-policy SAC	Dexterous manipulation and real-robot online learning	Adapted methods such as DSRL based on SAC or an off-policy critic

5.1 PPO + Locomotion

Typical task:

Teach a humanoid robot to climb stairs.

Common setup:

Isaac Lab + RSL-RL + PPO.

Main advantage:

Make full use of large-scale simulation environments.

5.2 SAC + Manipulation

Typical task:

A robot learns grasping and object manipulation through online interactions.

Main advantage:

Reuse experiences in the replay buffer and reduce the need to obtain new training data.

5.3 PPO-Style Methods + VLA

Typical task:

Use DPPO to post-train a Diffusion Policy with reinforcement learning.

Main advantage:

Directly improve the generative policy using policy gradients.

5.4 SAC / Off-policy Critic-Style Methods + VLA

Typical task:

Use DSRL to optimize a pretrained generative policy's latent variables, or use OGPO, Q-VGM, and similar methods to improve the generative policy.

Main advantage:

Improve data efficiency by using historical interaction experience.

An especially important distinction is:

OGPO and Q-VGM are off-policy generative-policy optimization approaches, but they are not equivalent to standard SAC.

---

6. How to Choose an RL Method for Your Task

The following table provides practical guidelines for project selection.

Project goal	First options to consider	Reason
Train a quadruped to walk from scratch	PPO	Mature parallel simulation workflows
Train motion control for humanoids such as G1	PPO	Established motion-control RL workflows
Train a dexterous hand to rotate objects in simulation	PPO	Parallel simulation can be exploited
Improve manipulation on a physical dexterous hand online	SAC or demonstration-initialized off-policy RL	Historical data can be reused
Fine-tune an existing Diffusion Policy	DPPO, OGPO	Direct improvement of the generative policy
Keep an existing Diffusion / Flow model unchanged	DSRL	Freeze the generator and optimize the noise policy
Improve VLA precision manipulation	Residual RL, RL Token	Train lightweight modules
Directly fine-tune a Flow Matching velocity field	Q-VGM and related methods	Update the generator using value-based signals
Avoid backpropagation through the full generation chain	Latent RL, Token RL, local Velocity Matching	Reduce online optimization cost

Practical choices should also consider model size, the availability of simulation, reward sparsity, and the budget for physical interactions and computation.

---

7. Typical Experimental Paths for the Two Research Directions

7.1 Robot Motion-Control RL

Recommended path:

```
Isaac Lab / MuJoCo
        |
        v
     PPO / SAC
        |
        v
 Train motion-control policy
```

```
        |
        v
 Domain Randomization
        |
        v
   Teacher-Student
        |
        v
     Sim-to-Real
        |
        v
Physical robot control
```

This route mainly focuses on:

How to make robots move more stably and flexibly, with stronger adaptability to physical environments.

Recommended starting points:

Isaac Lab + RSL-RL

7.2 Embodied Foundation-Model RL Post-Training

Recommended path:

```
       Robot Demonstrations
                |
                v
          Pretraining
                |
                v
     Diffusion / Flow / VLA
                |
                v
           RL Post-training
                |
     +----------+----------+
     |          |          |
     v          v          v
Latent RL Residual RL Full Fine-tuning
     |          |          |
   DSRL     RL Token   DPPO / OGPO
                           |
                           v
                    Velocity Field
                       Q-VGM
                |
                v
          Improved Robot Policy
                |
                v
           Real Robot Deployment
```

This route mainly focuses on:

How reinforcement learning can improve the task capabilities of pretrained foundation models with relatively little robot interaction data.

Recommended starting path:

DSRL → DPPO → OGPO → Q-VGM

---

8. Final Summary

For reinforcement learning in embodied AI, the field is best understood using the following three-level classification.

Classification level	Main categories	Core question
Level 1: Algorithm	PPO, SAC	How can environmental feedback improve a policy?
Level 2: Task	Locomotion, Manipulation, Foundation Model Post-training	Which robot task should RL solve?
Level 3: Optimization target	Low-level controller, Residual, Latent, Action Expert, Velocity Field	Which part of the policy does RL actually update?

In particular:

Robot motion-control reinforcement learning primarily follows the technical route of PPO, large-scale physics simulation, motion control, and Sim-to-Real.

Embodied foundation-model RL post-training builds on Diffusion / Flow / VLA models and focuses on improving pretrained policies with RL, including Latent RL, Residual RL, Token-Level RL, and Full Policy Fine-tuning.

The core distinction between them is:

- **Motion-control RL:** Primarily addresses how a robot acquires stable motion and manipulation control capabilities.
- **Foundation-model RL post-training:** Primarily addresses how to improve a model that already knows how to manipulate, further increasing success rate, speed, precision, and robustness.

Thus, embodied RL is not only a choice between PPO and SAC; it also requires choices about **task type, policy architecture, the source of training data, and the scope of policy updates.**